

ASSIGNMENT - REGRESSION ALGORITHM

Dataset: Insurance Charges Prediction

1. PROBLEM STATEMENT IDENTIFICATION

A client's requirements is to predict insurance charges based on several parameters.

justification:

Stage 1: Machine Learning: Training machine with historical data to make future predictions.

Stage 2: Supervised Learning: Dataset contains Input (age, bmi, etc) and Output (charges). Target column charges is provided to learn.

Stage 3: Regression: Target variable charges is a continuous numerical value (ex: 12345.67). We are predicting a price/value, not a category.

2. DATASET BASIC INFORMATION

Dataset Shape: (1338, 6)

Columns: age, sex, bmi, children, smoker, region, charges

Null Values: 0

numerical Features: age, bmi, children, charges

3. DATA PRE-PROCESSING

Handling Nominal Data: sex, smoker, region converted using One-Hot Encoding / Label Encoding.

Data Split: Train-Test Split (80% Training, 20% Testing), Random State = 0.

4. MODEL DEVELOPMENT (R2_Score Based)

I developed multiple models. Among them, Random Forest with Poisson criterion gave the highest R2 Score of 0.88389, so it was selected as the final model.

5.EXPERIMENT MY ALL THE RESEARCH

1.LINEAR REGRESSION =Default-0.78160

Standardisation = -0.791026

1.SUPPORT VECTOR MACHINE (SVM)

Kernal	C value	R2 score	Status
rbf	C=100	0.320031	Low
rbf	C=1000	0.810204	Good
Poly	C=100	0.617956	Moderate
Poly	C=1000	0.856648	Best in SVM
Sigmoid	C=10000	0.287471	Low/Not suitable

2.DECISION TREE

Criterion	Splitter	R2 score	Status
Squared_error	best	0.681819	Average
Friedman_mse	random	0.688960	Average

3.RANDOM FOREST – DETAILED TURNING

Parameters	R2 score	Status
Default, est=80	0.84983	Good
Squared_error, est=100	0.854778	Good
Friedman_mse ,est=100	0.854056	Good
Poisson,est=50	0.841910	Good
Poisson,est=100	0.852633	Good
Poisson,max_depth=None,est=100	0.85263	Good
Poisson,max_depth=None,est=50	0.8490	Good
Poisson,max_depth=5	0.879040	Very Good
Poisson,max_depth=6	0.867417	Very Good
Poisson, max_depth =5,est=100	0.8825	Excellent
Poisson, max_depth =5,est=1000	0.88389	OVERALL BEST – FINAL MODEL

6.FINAL MODEL AND JUSTIFICATION

My final model is **Random Forest with criterion='poisson', max_depth=5, n_estimators=1000** which gave **R2 Score = 0.88389**

Why I chose this:

1. Linear Regression (R2=0.78): This is a simple model. It cannot capture complex relationships between age, bmi, smoking and charges. So score is lowest. Rejected due to underfitting.

2. Decision Tree (R2=0.68): Score is only 0.68. Single tree overfits and is not stable. Rejected due to low accuracy.

3. SVM (Best R2=0.856): Even though it gave 0.85 in one trial, it is very unstable. With different C values, score went from -0.11 to +0.85. It is also very slow and sensitive to parameters. Rejected due to instability.

4. Random Forest (R2=0.88389): It gave highest score among all. More importantly, it is more stable than SVM - for all parameters (est=50, 100, 1000) it consistently gave 0.84+ score. With max_depth=5 it controls overfitting better. Hence selected as **FINAL MODEL**.